Thilak D

AI Data Scientist | Healthcare AI, LLMs and evaluation

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SUMMARY

AI data scientist building and evaluating healthcare AI in production. At Mondee, built the product's evals platform, analysed 50,000+ hospital encounters, fine-tuned a Telugu medical ASR model and ran a hospital pilot in which 88% of AI summaries were accepted without edits. First-author paper on hybrid knowledge-graph and vector retrieval (SAGE, ICCET 2025). MSc in Data Science and Computing; MBA in Strategy and Consulting in progress at BITS Pilani.

EXPERIENCE

Mondee Tech Private Limited (Mondee Tabhi Group) — Hyderabad

Chief of Staff (Data Science and AI) — Oct 2025 – Present

- Designed and built, with the data science team, the product's dedicated AI evaluation (evals) platform, which compares every AI-generated summary with the doctor-edited version and scores ASR transcripts and LLM outputs on metrics such as word error rate.
- The evaluation runs every release cycle, and the product team uses it to review the AI and engineering teams' work release by release.
- Analysed 50,000+ patient encounter records and 300+ doctor records (pre-visit AI assistant logs, doctor–patient transcripts, generated summaries and clinical decision support output); found a status bug where consultations put on hold were later marked finished with no record of whether they resumed, and drove the fix that took that error rate to near zero.
- Fine-tuned Google's MedASR into a Telugu medical speech-recognition model on 15,000 doctor–patient recordings (3–10 minutes each), and prepared datasets and training pipelines for Hindi, Tamil and Bengali. Benchmarked it against the commercial ASR in production (10.5% vs 7.5% word error rate; 4.2% vs 2.8% medical-term error) and, with latency also behind, kept the vendor and paused rollout: an evaluation-led build-versus-buy decision.
- Coordinate hospital pilots of the platform, now running across six hospitals including government institutions. Ran a one-month pilot end to end with 4 doctors and about 3,000 records, completed October 2026: 88% of AI summaries were accepted without edits, doctor feedback was positive, and record segregation came out as the main time cost to fix next.
- Built a live analytics dashboard (React, FastAPI) on doctor review data, now used at all six pilot hospitals: KPIs, a hospital timeline, a doctor leaderboard and a flagged-field triage board that routes issues to the AI, product and dev teams. Wrote its SOP, covering schema changes, seven API endpoints and a classification job.
- Lead a core non-technical implementation team of 6 that checks patient and doctor app flows on Android, iOS and web before release and drives fixes with the dev and AI teams. Run the release-cycle view leadership uses to track every function, across 3 release cycles in the past year.
- Prepared the platform for CDSCO licensing as a Class A AI medical device under the Medical Devices Rules, 2017: post-market surveillance plan, PSUR, ISO 13485 internal QMS audit and the MD-12 test licence application, now filed and at the review presentation stage.
- Built the capex, opex and three-year P&L model, with breakeven and IRR benchmarks, for a USD 100M AI-enabled care network of 4 inpatient hubs, each with 20 diagnostic spokes, plus mobile doctor and nurse units. The strategy is complete and now being implemented.

Sentinel AI ML Labs (SAIMA Labs) — Chennai (part-time)

Lead, AI ML Special Projects (AI Data Science Analyst) — May 2025 – Nov 2025

Junior Data Scientist — Jun 2024 – May 2025

- Built an AI dental-health prediction model and automated WhatsApp patient-engagement workflows for a dental hospital group.
- Built a Risk Assessment Score (RAS) bot for a startup foundation that rates a startup's investment potential from 20+ of the foundation's datasets, one startup at a time or in bulk from a CSV.
- Contributed to an AI cardiac-risk prediction model for a cardiac MedTech startup, working with cardiac specialists on early diagnosis.
- Led and mentored 5 interns across the RAS bot, product development and website projects.
- Helped pitch and present a cardiac MedTech startup's indigenous artificial-heart and ventricular-assist-device project at international innovation conferences in Colombo and Manila.

TaskBench (part of Sentinel AI ML Labs) — Chennai (internship)

Data Analyst (internship) — May 2024 – Jun 2024

- Built predictive and classification models in Python for market research, improving efficiency by 50%.
- Analysed 50,000+ live records and turned them into insights that shaped client marketing strategies.

- Prepared 2 due-diligence reports and 3+ proposal documents that supported decisions and client acquisition.

RESEARCH AND PROJECTS

SAGE (Semantic and Graph-based Engine): Hybrid Knowledge Retrieval for Ayushman Bharat Treatment Guidelines
Conference paper. Thilak D (first author), P Sunil Kumar, SVSN Sarma. Presented at the 13th International Conference on Contemporary Engineering and Technology (ICCET 2025), Chennai, Mar 2025. Paper ID ICCET250837.

- Fine-tuned LLaMA 3.1 (8B) and Phi-4 with Unsloth in 4-bit on 1,409 question–Cypher pairs, so plain-language questions become knowledge-graph queries; Phi-4 reached 0.86 BLEU and 0.73 exact match.
- Fused knowledge-graph and vector retrieval with weights set per query from answer relevance; the hybrid reached 0.95 answer relevance and 0.94 faithfulness, against 0.89 and 0.87 for vector-only RAG, and the best ROUGE-1 F1 of the three approaches (0.96).

Hybrid question answering with LangGraph: multi-hop RAG and knowledge graphs, MSc thesis, SSSIHL Jun 2024 – Apr 2025

- Built an agentic QA system in LangGraph for Ayushman Bharat questions, such as which private hospitals in a state perform a surgery and which documents to bring, with guardrails, query splitting, KG-or-RAG routing, multi-hop retrieval, ranking and answer fusion.
- Collected 150+ scheme documents with RPA and built a knowledge graph covering 50,000+ hospitals, 1,600+ surgeries and 9 states, alongside a chunked vector index of the guidelines.

Automated job recommendation system, SSSIHL Nov 2023 – Apr 2024

- Extracted required skills from job descriptions with Sentence Transformers, improving profile-matching accuracy by 25%.
- Ran semantic matching on Weaviate across 500+ listings and emailed personalised reports through Gmail.

EDUCATION

MBA, Strategy and Consulting Expected 2028
BITS Pilani, Work Integrated Learning Programme (for working professionals)

MSc, Data Science and Computing Jul 2023 – May 2025
Sri Sathya Sai Institute of Higher Learning (SSSIHL). CGPA 8.3. Thesis: hybrid question answering with LangGraph, multi-hop RAG and knowledge graphs.

BSc (Hons), Computer Science Jun 2020 – Jun 2023
Sri Sathya Sai Institute of Higher Learning (SSSIHL). CGPA 7.1

SKILLS

AI and ML: Machine learning, Deep learning, NLP, Predictive modelling, Speech recognition (ASR), Explainable AI (SHAP), Computer vision

LLMs and generative AI: RAG, Agentic AI (LangGraph), LangChain, Knowledge graphs, Ontology design, Hybrid vector and graph retrieval, LLM fine-tuning (Unsloth), Text-to-Cypher, LLM evaluation (BLEU, ROUGE, faithfulness), Small task-specific models

Programming: Python, SQL, R, JavaScript, TypeScript, C/C++, Java

Data and backend: pandas, NumPy, scikit-learn, PyTorch, TensorFlow, PySpark, Neo4j (Cypher), Weaviate, MongoDB, MySQL, FastAPI, Flask, AWS

Strategy and operations: Financial modelling (capex, opex, P&L, IRR), Market-entry strategy, CDSCO / MDR 2017, ISO 13485, ABDM / ABHA, Release management, Pitch decks, Power BI, Advanced Excel

ADDITIONAL INFORMATION

Certifications: Neo4j Certified Professional; AWS Academy Graduate: Cloud Foundations; Supervised Machine Learning: Regression and Classification; Databases and SQL for Data Science; Power BI

Honours: Best Intern 2024, Sentinel AI ML Labs; Ranked 15th nationally in Analytics Olympiad 2023, a machine learning hackathon hosted by MachineHack; Cleared the Actuarial Common Entrance Test (ACET), March 2023; House Championship Award, SSSIHL

Conferences: Presented paper, 13th International Conference on Contemporary Engineering and Technology (ICCET 2025), Chennai (Mar 2025); Presented and pitched, Global Innovation Summit and Retreat, Colombo, Sri Lanka (Sep 2025); Presented, ICE on I: PHILIPPiNEXT, 1st International Conference and Exposition on Innovations, Okada Manila, Parañaque City, Philippines (Jul 2025)

Leadership: Student President, Digital Enlightenment Club, SSSIHL (2023–2025); Co-founder, Eco-Einsteins Club, SSSIHL (2024–2025); Member, Institute Innovation Council, SSSIHL (2024); Captain, Annual Sports and Cultural Meet, SSSIHL (2025)

Languages: Tamil (native), English (fluent), Telugu (conversational)

Interests: Video editing, Music production, Reading tech blogs, Badminton, Cricket, Volleyball